for loop :

It is also **entry-controlled loop,** it is used to write concise code that means everything initialization, condition, and increment/decrement written in a single line.
It is preferable when number of iterations are **fixed**.

```
Syntax:   1              2                    4
for (initialization; condition; increment/decrement)
{
   // statements  3
}
```

For1.java

```
/* Print Welcome to Java 10 times */

void main()
{
  for(int i=1; i<=10; i++)
     {
        IO.println("Welcome to Java");
     }
}
```

For2.java

```
/* WAP print the number from 1 to n based on the user choice */

void main()
{
     int num = Integer.parseInt(IO.readln("Enter a number :"));

   for(int i=1; i<=num; i++)
     {
       IO.print(i+" ");

       if(i%5==0)
         {
        IO.println();
         }
     }
}
```

For3.java

```
/* WAP print all the even numbers from 1 to 100 */

void main()
{
     for(int i=2; i<=100; i=i+2)
     {
       IO.print(i+" ");
     }
}
```

For4.java

```
/* WAP print sum of first 100 natural number 1+2+3+4....100 */

void main()
{

   int sum = 0;
     for(int i=1; i<=100; i++)
     {
          sum += i;

     }
     IO.println("Sum of first 100 natural number is :"+sum);
}
```

For5.java

```
//use of break

void main()
{
     for(int i=1; i<=10; i++)
     {
      if(i==5)
               break;

           IO.println(i);
     }

}
```

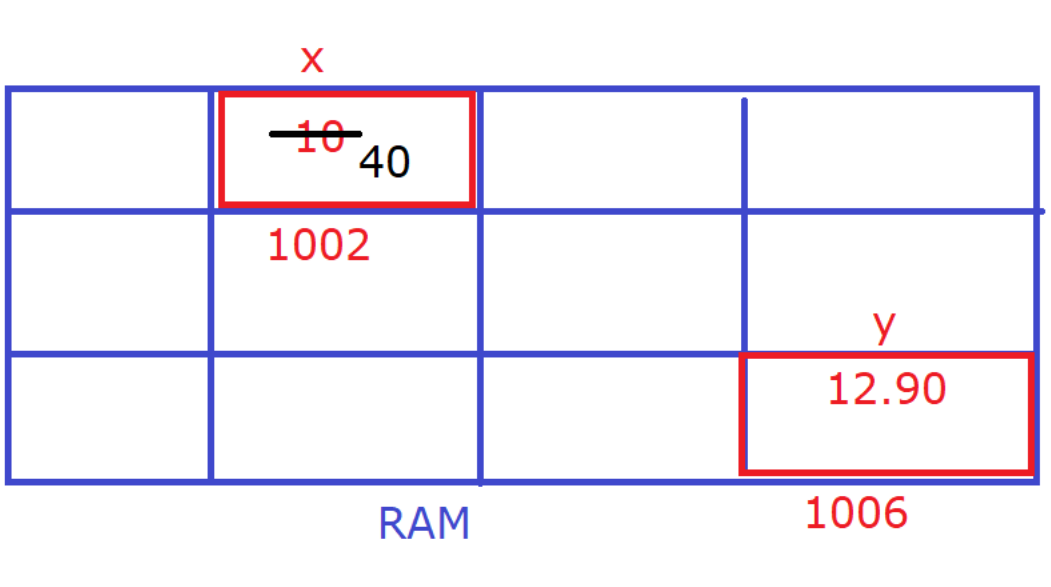
For6.java

```
//use of continue

void main()
{
     for(int i=1; i<=10; i++)
     {
      if(i==5)
               continue;  //Will skip the current iteration of loop

           IO.println(i);
     }

}
```

Variable :

* It is a name given for the memory location.
* It is used to hold some meaningful value.

```
Example : int x = 10;
          float y = 12.90f;
           x = x + 30;

           VARIABLE
           VARY + ABLE
           CHANGE + ABLE
```



* A variable can change its value during the execution of the program.

Drawback of an ordinary Variable :
----
1) An ordinary variable can hold only one value at a time in random memory location.

Example : int x = 10,20; //Invalid [In order to store multiple values in a sequential memory location we should use array.]

2) Once the program execution is completed we will not get back the value of the variable becuase they are stored in RAM i.e volatile memory. [File Handling, Here we will store the data in a file]

For each loop :

* It is an **enhanced for loop** which was introduced from JDK 1.5V.

* It is used to **fetch OR retrieve** the elements one by one from the **array OR collection**.

* Actually Java compiler will generate an ordinary for loop for this for each loop.

```
Syntax :
for(data type variable : array variable)
{
   //body of the for each loop
}
```

Test.java

```
void main()
{

    int []arr = {10,20,30,40,50};

    for(int x : arr)
    {
        IO.println(x);
    }
}
```

javac

Test.class

```
void main()
{

   int []arr = {10,20,30,40,50};

   for(int i=0; i<arr.length; i++)
   {
       IO.println(arr[i]);
   }
}
```

ForEach1.java

```
//for-each loop

void main()
{
     //Array Variable
     int []arr = {10,20,30,40,50};


     //Ordinary for loop
   for(int i=0; i<arr.length; i++)
   {
       IO.println(arr[i]);
   }

   IO.println("..............");

   //for each loop
     for(int x : arr)
     {
          IO.println(x);
     }


}
```

ForEach2.java

```
//for-each loop

void main()
{
     double []values = {12.90, 56.34, 23.90, 45.78};

     for(double value : values)
     {
     IO.println(value);
     }
}
```

values is an array variable which can hold multiple values, on the other hand value is an ordinary variable so, It can hold only one value for each iteration.

ForEach3.java

```
//for-each loop

void main()
{
     String []colors = {"White", "Blue", "Green", "Pink", "Red"};

     for(String color : colors)
     {
          IO.println(color);
     }
}
```

ForEach4.java

```
//for-each loop

void main()
{
     Object []values = {12, 45.89, 'A', true, "Java"};

     for(Object value : values)
     {
          IO.println(value);
     }
}
```